CHEM 210: Modern Applications of Chemistry

Discussion I: Stoichiometry Review

1.] Alums (formula is **$K[Al(SO_4)_2] \cdot 12\ H_2O$ (s)**, which contains 12 water molecules) are aluminum-based compounds that are sometimes found in shaving creams because they help reduce bleeding associated with minor cuts. The following is a balanced chemical equation for the production of alum. There are 12 water molecules attached to the salt in ALUM. We will discuss why these water molecules are present later in the course. Assume 298 K for temperature and 1 atm for pressure.

$$2\ Al(s) + 2\ KOH\ (aq) + 4\ H_2SO_4\ (aq) + 22\ H_2O\ (l) \rightarrow \underbrace{\mathbf{2\ K[Al(SO_4)_2] \cdot 12\ H_2O\ (s)}}_{\text{ALUM}} + 3\ H_2\ (g)$$

a) If 2.33 g of aluminum are reacted with 28 mL of 0.88 M KOH and 31 mL of 0.65 M H_2SO_4, what is the maximum volume of hydrogen that will be evolved in this chemical reaction? What mass, if any, of the aluminum remains?

b) If we need a minimum of yield of 4.5 kg of alum and are using soda cans as the source of aluminum, how many soda cans are needed to react to reach this yield? We are assuming an excess of the other reagents. Useful information:

Average weight of a can of soda (empty) 15.0 g

Mass Percent of Aluminum in the Can 93.7%

2.] A 2.747 gram sample of manganese metal is reacted with excess HCl gas to produce 3.22 L of H_2 (g) at 373 K and 0.951 atm and a manganese chloride $MnCl_x$. What is the formula of the manganese chloride compound?

3.] A person accidentally swallows a drop of liquid oxygen, O_2, and manages to avoid severe burns in the mouth and esophagus despite the extremely cold temperatures of the compound. However, the person is concerned about the volume to which the liquid will expand in the stomach once vaporized. Assume the drop has a volume of 0.050 mL, what volume of gas will be produced in the person's stomach at a body temperature of 37°C and a pressure of 1.0 atm? The density of liquid oxygen is 1.14 g/mL.

HINT: PV=nRT